

Ke Xiao

Sr./Staff Software Engineer, AI

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Summary

Senior Data Engineer with 12+ years of experience designing and delivering modern analytics platforms on Microsoft Azure, including Azure Data Factory, ADLS, Azure Databricks, Synapse Analytics, Azure SQL Database, and Fabric. Proven track record at ByteDance, Microsoft, Amazon, and VMware building metadata-driven data pipelines, dimensional models, and secure lakehouse architectures that enable high-quality, reliable, and scalable data solutions. Expert in advanced SQL, Spark, and Python with deep experience in API ingestion, JSON normalization, and data orchestration for both batch and near real-time workloads. Known for strong client leadership, clear communication, and mentoring teams while driving adoption of best-practice data strategies that improve business outcomes.

Skillsets

- 🚩 **Programming Languages:** SQL, Python, Scala, T-SQL, PySpark, DAX, Java, C#
- 🚩 **Data Engineering Processing:** Data Engineering, ETL/ELT Pipelines, Metadata-driven Orchestration, Dimensional Modeling, Star Schema Design, Data Lakehouse Architectures, Streaming Batch Processing, Data Quality Validation
- 🚩 **Azure Data Analytics:** Azure Data Factory, Azure Databricks, Azure Synapse Analytics, Azure Data Lake Storage (ADLS Gen2), Azure SQL Database, Microsoft Fabric, Azure Functions, Azure Event Hubs
- 🚩 **Databases Storage:** SQL Server, Azure SQL Managed Instance, Cosmos DB, Delta Lake, Parquet, PostgreSQL, Data Warehousing, Data Lakes
- 🚩 **Cloud Security:** Microsoft Azure, Azure DevOps, Infrastructure as Code (Bicep/Terraform), Azure Key Vault, Role-Based Access Control (RBAC), Data Security Governance, CI/CD for Data Pipelines, Monitoring Observability
- 🚩 **Integration APIs:** REST APIs, OAuth 2.0, JSON Normalization, API Ingestion Frameworks, Power Platform Integration, Event-driven Architectures, Message-based Integration, Webhooks
- 🚩 **Tools Practices:** Git, Agile Scrum, Technical Architecture Roadmapping, Stakeholder Management, Technical Documentation Diagramming, Mentoring Leadership, Performance Optimization, Testing Automation
- 🚩 **Tools Platforms:** Recall.ai, Statsig, Posthog, Mau, retool, Airtable, SendGrid, DataDog, Expo, Sentry, Deepgram, Auth0, StatusPage, Docker, Kubernetes, Redis, Open Search, Elasticsearch

Professional Experience

ByteDance

ASSOCIATES/STAFF (2024.3 - Present)

Bellevue, Washington, United States

- Led architecture of a unified Azure data lakehouse on ADLS, Databricks, and Synapse, consolidating analytics for multiple content platforms into governed domains.
- Implemented metadata-driven pipelines in Azure Data Factory and Databricks, enabling rapid onboarding of new data sources with reusable templates and configuration tables.
- Designed and optimized complex dimensional models and star schemas for creator analytics, improving Power BI query performance and report refresh times significantly.
- Built scalable Spark and PySpark jobs on Azure Databricks to process high-volume engagement events, supporting both batch and near real-time consumption patterns.
- Developed advanced SQL and T-SQL transformations in Synapse and Azure SQL Database, standardizing business logic and ensuring consistent KPIs across teams.
- Created a robust framework for ingesting REST APIs using OAuth 2.0, flattening nested JSON into normalized Delta Lake tables with automated schema evolution handling.
- Partnered with product and data science teams to define data contracts and SLAs, aligning engineering deliverables with ByteDance's fast-paced consumer app ecosystem.
- Implemented data quality checks, anomaly detection, and alerting using Databricks and Azure Monitor, reducing data incident resolution time and improving trust in metrics.
- Collaborated with security and governance teams to enforce RBAC, encryption, and PII handling policies across ADLS, Synapse, and Fabric workspaces for compliance.
- Mentored a distributed team of data engineers, conducting design reviews, pair programming, and workshops on lakehouse patterns and dimensional modeling best practices.
- Championed documentation and diagramming using Azure Architecture Center patterns, creating reusable blueprints for ingestion, curation, and serving layers.
- Worked closely with BI developers to optimize semantic models and DAX measures on top of Synapse and Fabric, ensuring low-latency dashboards for leadership decisions.
- Introduced CI/CD for data pipelines using Azure DevOps, implementing automated testing, validation, and environment promotion for ADF and Databricks assets.
- Partnered with regional business stakeholders to translate content monetization requirements into scalable Azure data solutions tailored to ByteDance's global markets.
- Led roadmap planning for the analytics platform, prioritizing initiatives that improved reliability, observability, and developer productivity across the data organization.

Microsoft

Senior Software Engineer, Data (2018.10 - 2024.2)

Greater Seattle Area

- Architected Azure-based data platforms for internal telemetry and customer-facing services, leveraging Data Factory, ADLS, and Synapse Analytics for end-to-end analytics.

- Designed dimensional models and star schemas for product usage analytics, enabling standardized reporting across multiple Microsoft 365 workloads and business units.
- Developed complex SQL and T-SQL stored procedures and views in Azure SQL Database, encapsulating transformation logic and ensuring reusable, testable data services.
- Implemented Spark and PySpark jobs on Azure Databricks to aggregate high-velocity diagnostic logs, supporting near real-time monitoring and anomaly detection scenarios.
- Built metadata-driven orchestration patterns in Data Factory, using parameterized pipelines, lookup-driven activities, and configuration tables to reduce duplication.
- Created reusable frameworks for ingesting REST APIs secured with OAuth 2.0, normalizing JSON payloads into relational structures and Delta Lake tables for analytics.
- Collaborated with Azure engineering and security teams to define data security architecture, combining data lake and warehouse patterns for governed self-service access.
- Led cross-functional design sessions with PMs, architects, and BI teams, translating business requirements into scalable Azure data solutions and clear technical designs.
- Introduced automated testing, validation, and deployment for data assets using Azure DevOps, improving reliability and shortening release cycles for analytics workloads.
- Mentored junior engineers and interns, providing guidance on SQL performance tuning, Spark optimization, and best practices for Azure data engineering projects.
- Partnered with Microsoft Consulting Services to support enterprise customers adopting Azure analytics, aligning reference architectures with real-world implementation feedback.
- Documented end-to-end data flows, lineage, and architecture diagrams, becoming a go-to resource for understanding telemetry pipelines across several Microsoft services.

Amazon

Software Development Engineer 2 (2017.10 - 2018.6)

Vancouver, British Columbia, Canada

- Designed and maintained large-scale data pipelines supporting retail and marketplace analytics, integrating transactional systems into a centralized analytics platform.
- Implemented ETL and ELT workflows using Spark and SQL to process high-volume order and traffic data, ensuring timely availability for reporting and forecasting.
- Developed dimensional models and star schemas for sales, inventory, and traffic domains, enabling consistent metrics across finance, operations, and category teams.
- Optimized complex SQL queries and partitioning strategies on columnar storage, significantly improving performance for peak-season reporting and dashboards.
- Collaborated with business intelligence engineers and product managers to translate Amazon retail requirements into robust data models and curated data sets.
- Introduced data quality checks, reconciliation processes, and monitoring dashboards, reducing data defects and increasing stakeholder confidence in analytics outputs.
- Documented data flows, lineage, and semantic definitions, helping new engineers and analysts quickly understand the retail analytics ecosystem and its dependencies.

VMware

Member Of Technical Staff (2014.4 - 2017.9)

Beijing, China

- Developed backend services and data integration components for VMware management products, focusing on reliable collection and storage of infrastructure telemetry.
- Implemented SQL-based data transformations and aggregation logic to support reporting on virtual machine performance, capacity, and configuration drift metrics.
- Collaborated with product managers and architects to design schemas and data access patterns that aligned with VMware's virtualization and cloud management offerings.
- Improved performance and scalability of data-heavy components through query optimization, indexing strategies, and efficient batching of telemetry ingestion.
- Created technical documentation and architecture diagrams for data flows within management products, supporting cross-team understanding and future enhancements.

Education

Beijing University of Posts and Telecommunications

Master's degree, Computer Science and Technology (2012 – 2014)

Central South University

Bachelor's degree, Computer Science and Technology (2007 – 2011)

Certification

- [DC: Skills for Success](#) by Dale Carnegie Training
- [Microsoft Global Hackathon 2022](#) by Microsoft
- [HTML, CSS, and JavaScript for Web Developers](#) by Coursera

Language

- ✓ Chinese – Native
- ✓ English – Fluent